OFFICIAL

# Epidural or Combined Spinal Epidural Analgesia in Labour Procedure

## 1. Scope

| Site | Operational Area | Applicable to |
|---|---|---|
| Armadale Health Service | Anaesthetics<br>Obstetrics | Medical<br>Nursing and Midwifery |

## 2. Purpose

The aim of this guideline is to set a standard for the safe provision of effective pain relief via epidural or Combined Spinal Epidural (CSE) for women in labour.

## 3. Equipment

- Sterile gown pack and gloves
- Sterile Epidural pack
- Epidural kit (Portex 18G or 16G, or CSE kit)
- 1% Lignocaine with Adrenaline (5ml ampoule)
- 10-20ml sterile normal saline
- Skin Asepsis
    - Unless contraindicated and where possible, commercially prepared swabs or swab sticks containing chlorhexidine are recommended for skin preparation before administering an epidural
    - If liquid chlorhexidine is used, it must be a dark tinted preparation as a visual cue that the liquid is not injectable
    - Skin preparation should be completed, and topical antiseptic discarded or handed off before exposing sterile equipment and injectable fluids on the sterile procedure area.
    - Non injectable fluids including chlorhexidine should never be decanted into an open container on a sterile procedure area.
    - After application the skin must be allowed to completely dry before the procedure is performed
- Sterile transparent film dressing (20 x 30cm or 15x8)
- Regional Anaesthesia Catheter Securement Device (“LockitTM) if requested
- Maternal and fetal monitoring equipment
- Epidural medications
    - 20mL premixed 0.125% bupivacaine with 5mcg/ml fentanyl
    - CADD pump

- CADD pump cassette OR prefilled bag of 0.0625% bupivacaine with 2.5mcg/mL fentanyl

## 4. Procedure

The anaesthetist (consultant or registrar) is responsible for insertion and management of the epidural or CSE in consultation with the Midwife / Specialist Obstetrician / General Practice Obstetrician (GPO). Insertion of the epidural may only be performed in the birthing suite.

- The anaesthetist must obtain consent from the patient for the epidural procedure
    - This can initially be verbal consent, but a written consent form must be signed once the patient is comfortable, which must be filed in the patient's medical record.
- Check for contraindications to insertion.

### 4.1 Contraindications for labour epidural / CSE

- Patient refusal, removal of consent or inability to co-operate
- Increased intracranial pressure
- Craniosacral anomalies (on a case-by-case basis)
- Skin or soft tissue infection at the site of needle insertion
- Untreated sepsis
- Coagulopathy
    - Thrombocytopenia associated with pre-eclampsia
        - A current platelet count (within 6 hours of epidural insertion or removal) must be available
        - Epidurals may be placed with platelet counts above 100
        - Epidurals may be placed with platelet counts between 70 and 100 provided coagulation studies are normal
    - Epidurals for patients with platelet levels less than 75 must be discussed with a consultant anaesthetist for a risk / benefit assessment
- Patients with pre-eclampsia should have recent platelets and coagulation studies (WITHIN 6 HOURS) performed both before epidural insertion and removal.
- Recent pharmacological anticoagulation
    - Not within 12 hours of prophylactic low molecular weight heparin
    - Not within 6 hours of prophylactic unfractionated heparin
- Thrombocytopenia due to other causes (e.g. gestational thrombocytopenia/ITP)
    - Considered on a case-by-case basis
- Uncorrected maternal hypovolaemia (e.g. haemorrhage, sepsis)

### 4.2 Preparation – prior to epidural / CSE insertion

- Take a relevant history and ensure recent vaginal examination prior to anaesthesia referral
- Obtain intravenous (IV) access using a16G cannula (18G minimum) with a running line preferably in the left hand or arm
- Check haematology as appropriate
- Obtain consent as above

## 4.3 Epidural catheter insertion – midwife role

- Provide care and support for the patient during the insertion of the epidural catheter
- Assist the patient with positioning for the epidural, either sitting or lying on their side, as requested by the anaesthetist
- Record maternal blood pressure and heart rate prior to epidural insertion
- Additional maternal monitoring may be indicated in certain situations, or at the request of the obstetrician or anaesthetist
- Continually monitor the fetal heart rate (FHR) according to EMHS Fetal Surveillance and Cardiotocography (CTG) Monitoring Policy during epidural insertion.
- Prepare the CADD pump
  - Set up and prime line
  - Program pump according to anaesthetist's prescription

## 4.4 Epidural catheter insertion – anaesthetist role

### 4.4.1 Neuraxial analgesia

**Initiation of analgesia:**

- Neuraxial analgesia may be initiated via an epidural or CSE technique
- Epidural:
  - "Premix" solution of 0.125% bupivacaine with fentanyl 5mcg/mL
  - 5 – 20mL in divided doses as needed
- CSE:
  - Intrathecal premix 1-2mL
  - Epidural premix 5-10mL as needed (the first dose down the epidural must be provided by the anaesthetist).

**Maintenance of analgesia:**

- Maintenance of analgesia is via the CADD Pumps
  - Keypad codes for CADD pumps can be found with the APS service and nurse manager
  - These codes should not be written anywhere that patients may be able to see them.
- The available premixed CADD solution is 0.0625% bupivacaine with 2.5mcg/mL fentanyl via the CADD pump bag, volume 100mL
- Programmed intermittent bolus and patient controlled epidural analgesia (PIB + PCEA)
  - Anaesthetist prescribes:
    - Programmed intermittent bolus: Default is 10mL every hour with a range of 0-12mL
    - PIB bolus interval: Default is 1 hour. The first dose is given forty minutes after initiation of the CADD pump. The CADD pump should be started before the IDC is inserted
    - PCEA dose: 5mL (range 0-10mL)
    - PCEA dose lockout interval: 15 minutes. NB: If a PCEA bolus has been requested by the patient within the lockout period (15mins) prior to the next PIB dose, the next programmed bolus will be withheld until the appropriate time interval has passed.
- PIB+PCEA orders should be checked by two midwives

  - On initial set-up of CADD pump
  - On change of shift
  - When changing the CADD cassette OR bag

#### 4.4.2 Rescue analgesia:

- Assess for other causes of failed block/breakthrough pain (e.g. uterine rupture, patient fully dilated, abruption, etc.)
- Check the block height bilaterally with ice to assess block
- Anaesthetist rescue boluses via the pump are possible, with a bolus volume between 0-20mL enabled
- Midwife initiated rescue:
  - Premix solution 0.125% bupivacaine with 5mcg/mL fentanyl 5-10mL; maximum of 10mL every hour, OR bupivacaine 0.25% 5mL 1hrly prn
  - Rectal pressure: 50mg pethidine or 50mcg fentanyl in 10mL normal saline
- Anaesthetist initiated rescue:
  - Consider pulling back catheter to 3-4cm in the space if the block is patchy or unilateral
  - No block:
    - 10-15mL 0.125% bupivacaine with 5mcg/mL fentanyl, OR
    - 5-10mL 0.25% bupivacaine (caution motor block), OR
    - 5-10mL 2% lignocaine (caution motor block)
  - Patchy block:
    - 10-20mL 0.0625% bupivacaine with 2.5mcg/mL fentanyl for an established but patchy block
    - Epidural clonidine 50-75mcg

#### 4.4.3 Instrumental delivery:

- Midwife or anaesthetist administered
  - 2% lignocaine with 1:200,000 adrenaline 5mL plus 5mL five minutes later, assessing block height between boluses with target T10 (umbilicus).

### 4.5 Assessment following epidural / CSE insertion and rescue analgesic top-ups

See also AKMR 178.1 Epidural / Spinal Analgesia Chart

#### 4.5.1 After initial dose of analgesic via epidural / CSE / rescue bolus:

- Assess and monitor the following:
  - Provide continuous fetal heart rate as per EMHS – [Fetal Surveillance and Cardiotocography (CTG) Monitoring Policy]
  - Blood pressure and pulse every 5 minutes for 20 minutes, then at 30 and 60 minutes
  - Pain score 2 hourly
  - Sensory levels (dermatomes) 2 hourly or sooner if required (e.g. pain, weakness)
- Midwife to insert urinary catheter as per [Bladder Management during labour and the postnatal period guideline] during maintenance of PIB + PCEA
- Assess and monitor the following:
  - Provide continuous fetal heart rate monitoring as per [Fetal Surveillance and Cardiotocography (CTG) Monitoring Policy]

- Blood pressure, heart rate, respiratory rate, conscious state and pain score hourly for four hours and then 2-hourly
- 4 hourly pain score with movement when awake
- Progression of labour as clinically indicated
- Maternal temperature 2 hourly while in labour
- Epidural catheter insertion site and tubing connection 12 hourly
- Monitor for pressure injuries and ensure pressure injury prevention strategies are in place
- Documentation on Epidural/Spinal Analgesia Chart: AKMR 178.1

### 4.5.2 Management of epidural related hypotension

- Call for assistance and notify the Duty Anaesthetist
- Stop CADD Pump and remove PCEA from the patient
- Give IV fluid bolus 250mL stat if patient has no cardiac contraindications or fluid overload risks (e.g. pre-eclampsia)
- Elevate legs (do not lower the head)
- Ensure the patient is positioned in the left lateral or left tilt position
- Administer oxygen via Hudson mask 6-8L/min
- Continuous electronic fetal monitoring

### 4.5.3 Management of high block (greater than T6)

**Prevention:**

- Ensure correct top-up technique
    - Check sensory block height prior to top-up
    - Review epidural insertion site to ensure catheter remains at same depth at skin
    - Administer a test dose prior to a treatment dose (e.g. 3 - 5mL of top-up solution)
    - Give the top-up dose slowly

**Management:**

- Call for assistance and notify the Duty Anaesthetist
- Stop CADD Pump and remove PCEA button from the patient
- If able, position the patient sitting up to allow block to “sink”, or in left lateral tilt position
- Administer oxygen via Hudson mask 6-8L/min and support the airway if required
- Assess fetal well-being

### 4.5.4 Ambulation after epidural / CSE analgesia in labour

**The patient may mobilise when:**

- The patient has scored a 1 on the Bromage Score in both legs
- Able to perform straight leg raise
- Able to flex knees with free movement of feet
- Sensation in the feet is described by the parturient as normal or near normal
- The supine and sitting blood pressure readings are within 15% of each other
- There is no dizziness on moving from sitting to standing

If the above criteria are met, the patient may ambulate only WITH TWO-PERSON ASSISTANCE and only in her room (the second person assisting may be a family member).

### 4.6 Discontinuing the epidural infusion / removal of epidural catheter

#### 4.6.1 Indications for epidural removal

- After the third stage of labour and when relevant, after the perineal repair is complete
- If signs of infection at insertion site are present (swelling, redness, pus) the Duty Anaesthetist should be notified PRIOR to epidural removal
- The patient's condition must be uncomplicated and stable with normal vital signs and no abnormal bleeding
- On removal, document the time and that the epidural tip is intact on the Epidural / Spinal Anaalgesia Chart AKMR 178.1 OR the anaesthetic chart AKMR 94
- Document wastage of the premixed epidural solution per the Medication Management procedure where additional documentation requirements apply for transparency and accountability, and to comply with legislation for the management of Schedule 8 medications.
    - Two persons that are authorised to handle S8/ S4R medications must be involved in the discard process
    - The quantity discarded must be recorded in the "amount discarded" column of the patient's MedX record.
    - This discard information will additionally be recorded on the epidural anaesthetic chart.
    - If a patient needs to be transferred out of labour suite (e.g. to operating theatres) the CADD pump should be disconnected and remaining analgesia solution discarded and documented on the patient's MedX record.
- NB. If thrombocytopenia, bleeding diatheses present, please contact the Duty Anaesthetist prior to removal

**CAUTION**:

Please check for administration of anticoagulation

**Low molecular weight heparin (LMWH)**: Removal of the catheter should occur at least 12 hours after the last dose of prophylactic LMWH and not within 4 hours of the next dose of prophylactic LMWH

**Unfractionated heparin (UFH):** Removal of the catheter should occur at least 6 hours after the last dose of prophylactic subcutaneous UFH and not within 2 hours of the next dose of prophylactic subcutaneous UFH.

**For doses of LMWH or UFH higher than normal, contact the Duty Anaesthetist prior to removing the epidural**

## 5. Policy Context

The following documents are required to give effect to this procedure

- Legislative
    - Medicine and Poisons Act 2014
    - Medicines and Poisons Regulations 2016

## 6. Supporting information

The following documents support the implementation of this procedure:

- HDWA
  - Disposal of medicines
- EMHS
  - Fetal Surveillance and Cardiotocography (CTG) Monitoring Policy
- AKG
  - Management of Thrombocytopenia in Pregnancy Guideline
  - Medication Policy
  - Medication Management Procedure
  - Bladder Management during labour and the postnatal period Guideline
- Other
  - WNHS Adult Medication Guideline Clonidine

## 7. Compliance, monitoring and evaluation

Compliance against this procedure will be evaluated with

- Monthly review of Datix CIMS aggregate data by the DTC for trending of medication-related errors.
- Multidisciplinary review of current medication systems which is coordinated by the pharmacy department once every four years, using the Medication Safety Self-Assessment (MSSA) tool provided by the ACSQHC.
- Quarterly retrospective audits of S8/ S4R registers and relevant paperwork, conducted by the safety and quality department.

## 8. Relevant standards

National Safety and Quality Health Service (NSQHS) Standards (second edition):

- Standard 4 – Action 4.03: Clinicians use organisational processes from the Partnering with Consumers Standard in medication management to:
- Standard 6 – Action 6.03: Clinicians use organisational processes from the Partnering with Consumers Standard to effectively communicate with patients, carers and families during high-risk situations

## 9. References

1. Chestnut DH, Wong CA, Tsen LC et al. Chestnut's Obstetric Anaesthesia: Principles and Practice, 5th Edition. Elsevier Saunders, Philadelphia 2014
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3. Capogna G, Stirparo S. Techniques for the maintenance of epidural analgesia in labor. Curr Opin Anesthesiol 2013; 26: 261-267
4. George RB, Allen TK, Habib AS. Intermittent Epidural Bolus Compared with Continuous Epidural Infusions for Labour Analgesia: A systematic review and meta- analysis. Anesth Analg 2013: 116(1): 133-144
5. Wong CA, McCarthy RJ, Hewlett B. The effect of manipulation of the programmed intermittent bolus time interval and injection volume on total drug use for labor epidural analgesia: A randomized controlled trial. Anesth Analg 2011; 112(4): 904-911

6. Breen TW, Shapiro T, Glass B, Foster-Payne D, Oriol NE. Epidural anaesthesia for labor in an ambulatory patient. Obstet Anesth 1993; 77: 919-924.

## 10. Policy owner

**Director Clinical Services**

Enquiries relating to this procedure may be directed to:

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## 11. Review

This procedure will be reviewed and evaluated as required to ensure relevance and currency. At a minimum it will be reviewed within one (1) year after first issue and at least every three (3) years thereafter.

| Version | Effective from | Effective to | Amendment(s) |
|---|---|---|---|
| V3 | 11 02 2026 | 11 02 2029 | Full review, significant changes throughout document |

## 12. Authorisation

This procedure has been authorised and issued by:

| | |
|---|---|
| **Authorised by** | Dr Tania Strickland - Director Clinical Services |
| **Authorisation date** | 09 02 2026 |
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